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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, ABRASIVE, EMERY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Emery cloth covered by this specification is a closed coat, jeans cloth backed coated abrasive in grit sizes as specified.

1.2 Classification. Emery cloth sheets shall be one of the following grit sizes as specified:

Sizes:

Fine.
Medium.
Coarse.
Extra Coarse.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards.

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification.

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Commerce Product Standard:

PS8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive.

3.1.1.1 Emery. The abrasive grain shall be blocky in shape in order to result in good polishing action and be free from extraneous material.

3.1.1.2 Grading, control and tolerance. Grading of abrasive grain shall be in accordance with PS8; and the tolerance therein shall apply to the abrasive grain recovered from the coated abrasive product.

3.1.2 Backing. The backing shall be jeans cloth for all grit sizes. The greige cloth shall conform to table I, after being conditioned at 70°F ± 2°F and 65 percent R.H. for at least 24 hours.

TABLE I. Backing requirements

Cloth type	Greige cloth Yards/Pound (1) + 5 percent	Thread count Per Inch (2) Warp	Filling
Jean	3.98	84	56

(1) Nominal yards per pound of greige cloth is on a 30-inch wide basis.
(2) A minus tolerance of three threads or a plus tolerance of five threads per inch will be permitted in the thread count of the greige cloth.

3.1.3 Adhesive. The adhesive shall be a good quality, animal hide glue base, or synthetic resins suitable for the purpose intended.

3.2 Coating.

3.2.1 Coating. The abrasive cloth shall be "closed coat". The abrasive grain shall be applied to one side of the cloth backing only, and the grain shall cover the entire surface of the backing.

3.3 Tensile breaking strength. The average breaking strength of the coated abrasive cloth shall be in accordance with table II, when tested as specified in 4.4.2.

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TABLE II. Tensile breaking strength

Grit size	Minimum average breaking strength	
	Direction of warp Pounds/Inch	Direction of filling Pounds/Inch
Fine, Medium, Coarse, and Extra Coarse	80	35

3.4 Form and dimensions.

3.4.1 Sheet. Unless otherwise specified (see 6.2), emery cloth shall be furnished in 9 x 11 inch sheets, with a tolerance of + 1/8 inch in length and + 1/16 inch in width, when measured at, and after conditioning at, 70° ± 2°F and 50 ± 2 percent relative humidity for at least 24 hours.

3.5 Identification marking. Each sheet of abrasive cloth shall be marked with the manufacturer's name or trademark (of such character that the source may readily be determined), and the grit size. All markings shall be legible and permanent.

3.6 Workmanship. Abrasive emery cloth shall be free from imperfections affecting its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105.

4.2.1 Sampling for examination of packaging and packing. Packaged and packed abrasive cloth in shipping containers shall be sampled in accordance with inspection level I, AQL - 4.0 percent defective.

4.2.2 Sampling for visual, dimensional, and workmanship characteristics of end item. Sampling for these characteristics shall be in accordance with inspection level S-4, AQL - 4.0 percent defective.

4.2.3 Sampling for tests. Samples for tests shall be selected in accordance with inspection level S-2. Samples may be selected from those selected in accordance with 4.2.2.

4.3 Examination and tests. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 Examination.

4.3.1.1 Packaged and packed abrasive cloth. Abrasive cloth, ready for shipment selected in accordance with 4.2.1, shall be examined with AQL's and classification of defects in accordance with MIL-P-3816.

4.3.1.2 Individual sheets. Samples of abrasive cloth selected in accordance with 4.2.2 shall be examined in accordance with table III. The AQL's for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE III. Classification of defects, end item

Examine (applicable paragraph)	Defects	Major	Minor
1.2	Size of grit not as specified or ordered.	X	
3.1.1.1	Extraneous material present in abrasive.	X	
3.1.2	Backing not as specified.	X	
3.2.1	Coating not completely covering surface of backing.	X	
3.4.1	Dimensions of sheets, not correct size or not within tolerances.	X	
3.5	Identification marking omitted; illegible, incomplete, or incorrect.		X
3.6	Loose grain.	X	
3.6	Sheets, torn or cracked.	X	
3.6	Damage or defects affecting function or serviceability.	X	

4.3.2 Acceptance tests. Samples selected in accordance with 4.2.3 shall be subjected to the tests in 4.4, AQL 10.0 percent defective.

4.4 Tests.

4.4.1 Grain size.

4.4.1.1 Grading for grit size. Grading to determine the grit size, including recovery of abrasive grain from the sheets, preparation of grain for grading, and procedures shall be in accordance with PS8.

4.4.2 Breaking strength. The tensile breaking strength of the coated abrasive, directly after conditioning for at least 24 hours at 70° + 2°F and 50 + 2 percent relative humidity, shall be determined by method 5102 of Federal Test Method Standard No. 191, except that the jaws of the testing machine shall be 5 inches apart, and that six specimens 1 inch wide shall be taken both longitudinally and transversely from sample sheets. The average tensile strength in both directions shall be computed from at least three specimens. Samples tearing at the jaws of the tensile testing machine should be discarded (see 3.3).

4.4.3 Certification of backing. The manufacturer shall furnish a certificate indicating that the greige cloth used in the backing complies with the requirements of paragraph 3.1.2. The certification shall contain test results performed on the same cloth manufactured with the same processes as the cloth used in the finished abrasive cloth.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing of abrasive cloth shall be in accordance with MIL-P-3816, as specified for level A, B, or C (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. Interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Emery cloth is a general purpose abrasive for cleaning and polishing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grit size (see 1.2).
(c) Form and dimensions required (see 3.4).
(d) Level of packaging and level of packing required (see 5.1).
(e) Civil or military marking as required (see 5.2).

6.3 Emery. The grain varies in color and is blocky in shape. This blockiness prevents deep penetration during use, resulting in low stock removal but a good polishing action. It is not recommended for use with woods.

6.4 Coating. A closed coat is one in which the abrasive grain is put on in such quantity as to entirely cover the backing.

6.5 Work value. Coated abrasives should be stored in a cool even temperature area where the relative humidity is between 40 to 50 percent. Do not store on concrete floors. The best work value will be obtained if stored and worked in rooms having a relative humidity of approximately 50 percent.

6.6 Abrasive grit size. The size of abrasive grain is indicated in table IV, which shows the approximate relation in size between Emery, Garnet, and artificial abrasives.

6.6.1 Cross reference for grit sizes. Product Standard PS8, Grading of Abrasive Grain on Coated Abrasive Products, uses the same classification for Garnet, Silicon Carbide and Aluminum Oxide abrasives. Table IV shows the comparative grit sizes. Commercial producers have also adopted this classification.

TABLE IV.

Grit sizes	Emery
240	
220	
180	Fine
150	
120	Medium
100	
80	Coarse
60	Extra Coarse
50	
40	
36	
30	

PREPARING ACTIVITY: GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of specification to obtain extra copies and other documents referenced herein.

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